



Lab 3 - Complete Codebase & File Staging Guide (Sanitized)

This guide provides the **exact code you need to add**, organized by file location, with dependencies mapped. It builds directly on your Lab 2 infrastructure without breaking anything.

Pre-Stage: File Structure & Location Map

Before writing any code, set up your folder structure. This is critical for multi-region Terraform.

Recommended Directory Structure

```
lab-3/  
├── tokyo/                                # ap-northeast-1 (Data Au  
thority)  
│   └──
```

Dependency Chain (Execute in This Order)

This is the critical path. Do NOT skip ahead.

```
|  
├──  
|   PHASE 1: TOKYO FOUNDATION (Must complete before São Paulo)  
|  
├──  
└──  
|   1.1 02-providers.tf → ADD aws.saopaulo provider alias
```

1.2	03-variables.tf	→ ADD	tokyo_vpc_cidr, saopaulo_vpc_cidr,
			saopaulo_tgw_id, enable_tgw
1.3	04-3a-tokyo-tgw.tf	→ NEW:	shinjuku_tgw01 + VPC attachment
			(attaches to helga_vpclab2a)
1.4	05-outputs.tf	→ ADD	tokyo_tgw_id, tokyo_vpc_cidr_block
1.5	terraform apply	→ Creates	TGW, captures tokyo_tgw_id



PHASE 2: SÃO PAULO DEPLOYMENT (Separate Terraform State)

2.1	provider.tf	→ NEW:	aws provider for sa-east-1
2.2	variables.tf	→ NEW:	saopaulo CIDRs, tokyo_vpc_cidr,
			tokyo_tgw_id, tokyo_rds_endpoint
2.3	main.tf	→ NEW:	liberdade_vpc01, subnets, NAT, routes
		⚠	NO RDS - APPI COMPLIANCE
2.4	sao-paulo-tgw.tf	→ NEW:	liberdade_tgw01 + VPC attachment
2.5	outputs.tf	→ NEW:	saopaulo_tgw_id, saopaulo_

vpc_cidr |
| 2.6 terraform apply → Creates São Paulo infra, captures saopaulo_tgw_id |
|
|



| PHASE 3: TOKYO COMPLETION (After São Paulo TGW exists) |
|
|

| 3.1 terraform.tfvars → ADD saopaulo_tgw_id = "tgw-xxx" (from Phase 2) |
|

| 3.2 04-3a-tokyo-tgw.tf → ADD shinjuku_to_liberdade_peer01 (peering) |
|

| ADD shinjuku_route_to_liberdade (TGW route) |
|

| 3.3 04-3b-tokyo-routes.tf → NEW: aws_route in helga_private_rt01 |
|

| (destination: 10.1.0.0/16 → TGW) |
|

| 3.4 04-3c-rds-sg-update.tf → NEW: aws_security_group_rule |
|

| (helga_rds_sg01 allows 10.1.0.0/16 on 3306) |
|

| 3.5 terraform apply → Creates peering (state: pending Acceptance) |
|
|



| PHASE 4: SÃO PAULO FINALIZATION (Accept Peering) |

```
|
|-----|
| 4.1 sao-paulo-tgw.tf → ADD data source for pending peering
ring
|
|                                     ADD liberdade_accept_peer01 (a
ccepter)
|
|                                     ADD liberdade_route_to_shinjuku
(TGW route)
| 4.2 sao-paulo-routes.tf → NEW: aws_route in liberdade_private_rt01
|
|                                     (destination: 10.0.0.0/16 → TGW)
| 4.3 terraform apply → Accepts peering (state: available)
|
|-----|
```



PHASE 5: VERIFICATION & AUDIT (Lab 3B)

- 5.1 Verify RDS in Tokyo ONLY (no RDS in sa-east-1)
- 5.2 Verify TGW peering state = "available"
- 5.3 Verify route tables have cross-region routes
- 5.4 Verify helga_rds_sg01 allows 10.1.0.0/16
- 5.5 Test: São Paulo EC2 → Tokyo RDS connectivity
- 5.6 Run Python evidence scripts

5.7 Assemble audit-pack/ folder

CIDR Blocks Reference



CRITICAL: Before proceeding, confirm your CIDR blocks. These must not overlap.

Region	Resource	CIDR Block
Tokyo	VPC	10.0.0.0/16
Tokyo	Public Subnets	10.0.1.0/24 , 10.0.2.0/24
Tokyo	Private Subnets	10.0.11.0/24 , 10.0.12.0/24
São Paulo	VPC	10.1.0.0/16
São Paulo	Public Subnets	10.1.1.0/24 , 10.1.2.0/24
São Paulo	Private Subnets	10.1.11.0/24 , 10.1.12.0/24

Quick Reference: Resource Naming

Region	Prefix	Example
Tokyo	shinjuku_	shinjuku_tgw01
São Paulo	liberdade_	liberdade_tgw01

Key Integration Points with Lab 2 Code

Existing Lab 2 resources that Lab 3 connects to:

Lab 2 Resource	Reference Name	File Location
Tokyo VPC	aws_vpc.helga_vpclab2a	04-2ab-Main.tf

Lab 2 Resource	Reference Name	File Location
Tokyo Private Subnets	<code>aws_subnet.helga_private_subnets</code>	04-Main.tf (Lab 1C)
Tokyo RDS	<code>aws_db_instance.helga_rds01</code>	04-Main.tf (Lab 1C)
Tokyo RDS Security Group	<code>aws_security_group.helga_rds_sg01</code>	04-Main.tf (Lab 1C)
Private Route Table	<code>aws_route_table.helga_private_rt01</code>	04-Main.tf (Lab 1C)

Phase Subpages

See subpages below for detailed code for each phase.

- [1 Phase 1 - Tokyo Foundation Code \(Sanitized\)](#)
- [2 Phase 2 - São Paulo Deployment Code \(Sanitized\)](#)
- [3 Phase 3 - Tokyo Completion Code \(Sanitized\)](#)
- [4 Phase 4 - São Paulo Finalization Code \(Sanitized\)](#)
- [5 Phase 5 - Verification & Audit Pack \(Sanitized\)](#)
- [6 Phase 6 - Audit Evidence Pack Assembly \(Sanitized\)](#)